

Table 6: Comparison of DFPRP with and without restart

N	Pdim		DFPRP1(with restart)				DFPRP2(without restart)			
			IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM
1	5000	x_a^1	1	3	0.0081	0	1	3	0.0784	0
	5000	x_a^2	1	3	0.0128	0	1	3	0.0273	0
	5000	x_a^3	1	3	0.0063	0	1	3	0.0178	0
	5000	x_a^4	1	3	0.0060	0	1	3	0.0076	0
	5000	x_a^5	1	3	0.0073	0	1	3	0.0081	0
	5000	x_a^6	1	3	0.0064	0	1	3	0.0083	0
	10000	x_a^1	1	3	0.0089	0	1	3	0.0099	0
	10000	x_a^2	1	3	0.0087	0	1	3	0.0101	0
	10000	x_a^3	1	3	0.0097	0	1	3	0.0120	0
	10000	x_a^4	1	3	0.0080	0	1	3	0.0102	0
	10000	x_a^5	1	3	0.0083	0	1	3	0.0103	0
	10000	x_a^6	1	3	0.0127	0	1	3	0.0098	0
	50000	x_a^1	1	3	0.0266	0	1	3	0.0237	0
	50000	x_a^2	1	3	0.0242	0	1	3	0.0270	0
	50000	x_a^3	1	3	0.0519	0	1	3	0.0261	0
	50000	x_a^4	1	3	0.0545	0	1	3	0.0253	0
	50000	x_a^5	1	3	0.0237	0	1	3	0.0347	0
	50000	x_a^6	1	3	0.0215	0	1	3	0.0264	0
2	5000	x_a^1	1	3	0.0056	0	1	3	0.0071	0
	5000	x_a^2	1	4	0.0071	0	1	4	0.0090	0
	5000	x_a^3	1	3	0.0069	2.22E-16	1	3	0.0070	2.22E-16
	5000	x_a^4	3	6	0.0102	0	4	9	0.0167	5.21E-11
	5000	x_a^5	1	3	0.0060	0	1	3	0.0072	0
	5000	x_a^6	3	6	0.0103	0	4	9	0.0175	5.26E-11
	10000	x_a^1	1	3	0.0073	0	1	3	0.0110	0
	10000	x_a^2	1	4	0.0087	0	1	4	0.0118	0
	10000	x_a^3	1	3	0.0078	2.22E-16	1	3	0.0099	2.22E-16
	10000	x_a^4	3	6	0.0176	0	4	9	0.0378	7.38E-11
	10000	x_a^5	1	3	0.0074	0	1	3	0.0091	0
	10000	x_a^6	3	6	0.0141	0	4	9	0.0217	7.41E-11
	50000	x_a^1	1	3	0.0177	0	1	3	0.0208	0
	50000	x_a^2	1	4	0.0244	0	1	4	0.0287	0
	50000	x_a^3	1	3	0.0208	2.22E-16	1	3	0.0206	2.22E-16
	50000	x_a^4	3	6	0.0502	0	4	11	0.0856	7.77E-11
	50000	x_a^5	1	3	0.0225	0	1	3	0.0237	0
	50000	x_a^6	3	6	0.0446	0	4	11	0.0854	7.78E-11
3	5000	x_a^1	1	4	0.0073	0	1	4	0.0092	0
	5000	x_a^2	3	8	0.0132	0	7	20	0.0359	0
	5000	x_a^3	5	12	0.0236	4.44E-16	2	6	0.0115	4.44E-16
	5000	x_a^4	1	4	0.0068	0	1	4	0.0072	0
	5000	x_a^5	1	4	0.0071	0	1	4	0.0188	0
	5000	x_a^6	1	4	0.0114	0	1	4	0.0081	0
	10000	x_a^1	1	4	0.0088	0	1	4	0.0096	0
	10000	x_a^2	3	8	0.0185	0	6	18	0.0462	0
	10000	x_a^3	5	12	0.0278	4.44E-16	2	6	0.0157	4.44E-16
	10000	x_a^4	1	4	0.0083	0	1	4	0.0114	0
	10000	x_a^5	1	4	0.0099	0	1	4	0.0117	0
	10000	x_a^6	1	4	0.0095	0	1	4	0.0282	0
	50000	x_a^1	1	4	0.0260	0	1	4	0.0313	0
	50000	x_a^2	3	8	0.0768	0	6	18	0.1663	0
	50000	x_a^3	5	12	0.1099	4.44E-16	2	6	0.0538	4.44E-16
	50000	x_a^4	1	4	0.0280	0	1	4	0.0326	0
	50000	x_a^5	1	4	0.0291	0	1	4	0.0424	0
	50000	x_a^6	1	4	0.0643	0	1	4	0.0318	0

Table 7: Comparison of DFPRP with and without restart

N	Pdim		DFPRP1(with restart)				DFPRP2(without restart)			
			IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM
4	5000	x_a^1	1	2	0.0058	0	1	2	0.0076	0
	5000	x_a^2	2	3	0.0088	0	2	3	0.0096	0
	5000	x_a^3	6	9	0.0183	2.22E-16	4	7	0.0176	3.14E-16
	5000	x_a^4	5	8	0.0167	7.78E-23	4	7	0.0164	4.07E-12
	5000	x_a^5	2	3	0.0188	0	2	3	0.0100	0
	5000	x_a^6	5	8	0.0164	7.79E-23	4	7	0.0168	4.08E-12
	10000	x_a^1	1	2	0.0076	0	1	2	0.0110	0
	10000	x_a^2	2	3	0.0142	0	2	3	0.0193	0
	10000	x_a^3	6	9	0.0322	3.14E-16	4	7	0.0247	3.14E-16
	10000	x_a^4	5	8	0.0265	2.82E-23	4	7	0.0269	4.96E-12
	10000	x_a^5	2	3	0.0133	0	2	3	0.0139	0
	10000	x_a^6	5	8	0.0316	2.82E-23	4	7	0.0282	4.97E-12
	50000	x_a^1	1	2	0.0210	0	1	2	0.0243	0
	50000	x_a^2	2	3	0.0472	0	2	3	0.0430	0
	50000	x_a^3	6	9	0.1741	3.14E-16	4	7	0.1418	3.14E-16
	50000	x_a^4	5	8	0.1587	2.59E-24	4	7	0.0860	9.80E-12
	50000	x_a^5	2	3	0.0685	0	2	3	0.0435	0
	50000	x_a^6	5	8	0.1857	2.60E-24	4	7	0.0903	9.80E-12
5	5000	x_a^1	1	3	0.0068	0	1	3	0.0086	0
	5000	x_a^2	1	3	0.0132	0	1	3	0.0079	0
	5000	x_a^3	1	3	0.0059	0	1	3	0.0067	0
	5000	x_a^4	1	3	0.0149	0	1	3	0.0072	0
	5000	x_a^5	1	3	0.0088	0	1	3	0.0083	0
	5000	x_a^6	1	3	0.0145	0	1	3	0.0061	0
	10000	x_a^1	1	3	0.0193	0	1	3	0.0092	0
	10000	x_a^2	1	3	0.0139	0	1	3	0.0147	0
	10000	x_a^3	1	3	0.0106	0	1	3	0.0120	0
	10000	x_a^4	1	3	0.0077	0	1	3	0.0116	0
	10000	x_a^5	1	3	0.0133	0	1	3	0.0096	0
	10000	x_a^6	1	3	0.0200	0	1	3	0.0091	0
	50000	x_a^1	1	3	0.0322	0	1	3	0.0263	0
	50000	x_a^2	1	3	0.0254	0	1	3	0.0272	0
	50000	x_a^3	1	3	0.0574	0	1	3	0.0287	0
	50000	x_a^4	1	3	0.0301	0	1	3	0.0262	0
	50000	x_a^5	1	3	0.0446	0	1	3	0.0472	0
	50000	x_a^6	1	3	0.0500	0	1	3	0.0254	0
6	5000	x_a^1	6	13	0.0627	1.85E-11	6	13	0.0327	1.85E-11
	5000	x_a^2	6	13	0.0495	1.49E-11	6	13	0.0340	1.49E-11
	5000	x_a^3	6	13	0.0616	2.27E-11	6	13	0.0343	2.26E-11
	5000	x_a^4	6	13	0.0661	1.86E-11	6	13	0.0331	1.86E-11
	5000	x_a^5	6	13	0.0535	1.86E-11	6	13	0.0324	1.86E-11
	5000	x_a^6	6	13	0.0348	1.86E-11	6	13	0.0341	1.86E-11
	10000	x_a^1	6	13	0.0705	2.61E-11	6	13	0.0492	2.62E-11
	10000	x_a^2	6	13	0.0693	2.10E-11	6	13	0.0536	2.11E-11
	10000	x_a^3	6	13	0.0974	3.20E-11	6	13	0.0506	3.20E-11
	10000	x_a^4	6	13	0.0853	2.63E-11	6	13	0.0536	2.63E-11
	10000	x_a^5	6	13	0.0593	2.63E-11	6	13	0.0512	2.63E-11
	10000	x_a^6	6	13	0.0924	2.63E-11	6	13	0.0533	2.63E-11
	50000	x_a^1	6	13	0.2070	5.83E-11	6	13	0.2254	5.84E-11
	50000	x_a^2	6	13	0.2401	4.71E-11	6	13	0.2395	4.70E-11
	50000	x_a^3	6	13	0.2063	7.15E-11	6	13	0.2352	7.15E-11
	50000	x_a^4	6	13	0.2066	5.88E-11	6	13	0.2276	5.88E-11
	50000	x_a^5	6	13	0.3509	5.87E-11	6	13	0.2306	5.87E-11
	50000	x_a^6	6	13	0.2312	5.88E-11	6	13	0.2367	5.88E-11

Table 8: Comparison of DFPRP with and without restart

N	Pdim		DFPRP1(with restart)				DFPRP2(without restart)			
			IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM
7	5000	x_a^1	8	15	0.0685	2.22E-12	153	309	0.6796	9.11E-11
	5000	x_a^2	10	18	0.0431	6.04E-11	134	272	0.5858	9.30E-11
	5000	x_a^3	13	24	0.0645	6.16E-12	156	316	0.7082	9.35E-11
	5000	x_a^4	17	30	0.1379	1.91E-12	137	278	0.6091	9.82E-11
	5000	x_a^5	13	22	0.0943	8.99E-11	148	300	0.6210	9.90E-11
	5000	x_a^6	17	30	0.0971	1.75E-12	134	272	0.5892	9.79E-11
	10000	x_a^1	8	15	0.0972	3.14E-12	156	315	0.8917	8.94E-11
	10000	x_a^2	10	18	0.0897	8.54E-11	137	278	0.8371	9.13E-11
	10000	x_a^3	17	30	0.1113	7.30E-12	159	322	0.9580	9.18E-11
	10000	x_a^4	17	30	0.1088	2.64E-12	134	272	0.8376	9.42E-11
	10000	x_a^5	13	24	0.1484	9.78E-13	151	306	0.9052	9.72E-11
	10000	x_a^6	17	30	0.1230	2.53E-12	103	211	0.6885	9.97E-11
	50000	x_a^1	8	15	0.1802	7.01E-12	162	327	3.2147	9.63E-11
	50000	x_a^2	10	20	0.2601	1.47E-12	143	290	2.8776	9.83E-11
	50000	x_a^3	18	30	0.4465	5.02E-11	165	334	3.2787	9.88E-11
	50000	x_a^4	17	30	0.3505	5.81E-12	140	284	2.9463	8.92E-11
	50000	x_a^5	13	24	0.2857	2.19E-12	158	320	3.1408	9.27E-11
	50000	x_a^6	17	30	0.3637	5.76E-12	136	276	2.7900	9.12E-11
8	5000	x_a^1	11	23	0.0502	8.65E-11	12	27	0.0661	6.68E-11
	5000	x_a^2	11	25	0.0723	8.60E-11	12	25	0.0628	4.51E-11
	5000	x_a^3	12	25	0.0583	4.63E-11	13	27	0.0647	4.08E-11
	5000	x_a^4	11	25	0.0575	7.91E-11	12	27	0.0611	8.41E-11
	5000	x_a^5	11	23	0.0504	7.22E-11	12	27	0.0625	6.55E-11
	5000	x_a^6	11	25	0.1239	4.86E-11	12	27	0.0654	6.77E-11
	10000	x_a^1	12	27	0.1169	6.04E-11	13	27	0.0944	4.01E-11
	10000	x_a^2	11	23	0.0833	6.57E-11	13	27	0.1106	3.50E-11
	10000	x_a^3	12	27	0.0943	3.16E-11	12	27	0.1324	6.90E-11
	10000	x_a^4	12	27	0.1117	8.13E-11	13	27	0.1126	2.69E-11
	10000	x_a^5	12	27	0.0980	6.58E-11	13	27	0.1363	3.92E-11
	10000	x_a^6	12	27	0.1276	7.45E-11	13	27	0.1144	3.78E-11
	50000	x_a^1	14	29	0.4438	3.13E-11	14	29	0.5000	9.75E-11
	50000	x_a^2	13	29	0.5672	9.21E-11	13	29	0.4484	8.04E-11
	50000	x_a^3	14	29	0.4273	3.77E-11	15	31	0.5254	2.01E-11
	50000	x_a^4	14	29	0.4879	4.56E-11	14	29	0.5304	7.62E-11
	50000	x_a^5	14	29	0.4254	3.69E-11	14	29	0.7596	9.80E-11
	50000	x_a^6	14	29	0.6326	4.24E-11	14	31	0.4776	4.50E-11